Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Capstone hypothesis and baseline: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Write a testable capstone **hypothesis** and a credible **baseline**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Documenting a network experiment

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Record a network experiment with enough detail to replay its calculations

Reference example, not the only acceptable wording: Teacher explanation: Predict max(0, a*x); start a = 1 with no bias; rate 0.1; two full-batch updates; MSE is mean squared **prediction** error.

T1 and T2 train, V1 validates; no shuffling or normalisation. Select lower validation MSE, ties to earlier update. Record the actual Python version, alongside the code and Python float arithmetic.

Dataset shift in neural models

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain how a changed **input** mix can change overall accuracy

Reference example, not the only acceptable wording: Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both.

Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim. The mix changes while observed group rates remain the same.

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate vector length and normalise a nonzero vector

Reference example, not the only acceptable wording: Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8].

The length difference between Q and A disappears; the opposite direction of C is preserved.

For the compact alternative schedule, use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in **training** and **evidence**, not a real-world deployment.

Preserve written plans, data roles and raw **prediction**s. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

Outcomes

Write a testable capstone **hypothesis** and a credible **baseline**

Specify data roles and a selection rule before inspecting results

Calculate **baseline** MSE and distinguish a **success criterion** from a result

Preparation

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

Prerequisite answer

Validation influences selection, so its score is not independent final **evidence**. Freeze the chosen procedure before opening the final test.

End goal for the pictorial case

Explain capstone **hypothesis** and **baseline** with a traceable toy-network case and a bounded conclusion.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Capstone hypothesis and baseline using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Capstone hypothesis and baseline · 1

A capstone needs a question that **evidence** can answer. "Build a better network" leaves better undefined.

Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

Explain Capstone hypothesis and baseline · 2

The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

A stronger comparison may also include an ordinary fitted line.

Explain Capstone hypothesis and baseline · 3

Training rows determine updates. Validation rows compare the prespecified candidate with the **baseline**. The final-test **input**s and targets remain sealed until the procedure and reporting plan are frozen.

MSE = sum of squared **prediction** errors divided by the number of evaluated rows. **Baseline prediction**s on V1 and V2 are 3 and 4. Errors are -3 and -4, squared errors 9 and 16, and validation MSE is 25/2 = 12.5.

There are two validation cases.

Term	Meaning
Hypothesis	A testable expectation with a stated comparison and outcome
Baseline	A specified reference procedure evaluated on the same cases
Success criterion	A rule written before evaluation for judging the measured result
Reserved test	Cases kept out of fitting, tuning and selection

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Guided: errors -3 and -3 give MSE 9 on two rows. Candidate **prediction**s are missing, so the **hypothesis** remains untested.

Exit: final rows became selection **evidence**; document the change, freeze the revised plan and reserve new independent final cases.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

The validation criterion is candidate MSE strictly below **baseline** MSE on V1 and V2. Equality does not pass. The final criterion is lower MSE than the same **baseline** on the sealed final rows.

Report both scores and every raw **prediction** whether or not the criterion passes. Record the rule, **training** rows, update count, learning rate, loss, data versions and code version.

A criterion specifies how to judge a future result; it is not **evidence** that the result already happened.

Keep validation and final criteria separate. Do not silently add epochs or change pre**process**ing after seeing a weak result. A changed plan is a new version and needs an honest account of what **evidence** informed it.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone hypothesis and baseline

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

The facts we will use

Supplied facts

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

AI ACADEMY / AI-5/33

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the prediction

B should have more correct labels than A on the same eight patterns.



2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.



3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the prediction	B should have more correct labels than A on the same eight patterns.
2. Declare the baseline	A is a documented corner-only network, not a deliberately broken version.
3. Reserve evaluation	Keep the eight patterns out of model selection and tuning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

What this example does not establish

Keep the stated limit

Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Change one thing in the pictured case

Change one thing

Change: Give only B the eight evaluation labels during **training**.

Predicted result

Prediction: The proposed comparison becomes contaminated rather than a valid held-out test.

Why the result follows

Reason: One version has gained privileged information from the reserved evaluation set.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Capstone hypothesis and baseline, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and ask students to explain the data roles.
7 to 18 minutes	Read section 1 and complete W1 with a recorded reason.
18 to 32 minutes	Trace section 2 and complete W2; compare raw values before conclusions.
32 to 45 minutes	Explain the diagram and complete W3 in pairs; record any support.
Session 2 start to 8 minutes	Complete the guided response and discuss why the evidence has limits.
8 to 20 minutes	Read the remaining explanation and prepare a concise experiment record.
20 to 35 minutes	Collect W4 independently. Require calculations and a bounded claim.
35 to 45 minutes	Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Reduce help deliberately

Completion practice

Completion task: Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Reasoning to listen for

Teacher check: Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Reasoned teacher answer

The change is adding the centre feature; A is corner-only. The outcome is correct labels out of the same eight patterns. Using reserved labels to train B leaks evaluation information and invalidates the unseen comparison.

Changed-case reasoning: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Capstone hypothesis and baseline investigation and use the first unsupported step to choose a teaching response.

A **hypothesis** is a result: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

A weak **baseline** proves neural networks help: An initial-weight reference checks improvement from that starting point, not superiority to a fitted line.

Renaming a used test restores independence: A new name cannot undo using its results for selection.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Capstone hypothesis and baseline questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: errors -3 and -3 give MSE 9 on two rows. Candidate **prediction**s are missing, so the **hypothesis** remains untested.

Exit: final rows became selection **evidence**; document the change, freeze the revised plan and reserve new independent final cases.

W1 Write the hypothesis

Workbook question · W1

State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

Teacher reasoning · W1

Teacher explanation: Predict max(0, a*x) without bias; **baseline** a = 1; candidate starts at 1, learning rate 0.1, two full-batch updates on T1/T2.

Hypothesis: candidate validation MSE on V1/V2 is strictly below the **baseline** MSE.

W2 Calculate the reference score

Workbook question · W2

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Teacher reasoning · W2

Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

W3 Audit the plan

Workbook question · W3

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

Teacher reasoning · W3

Teacher explanation: It omits architecture, **baseline**, **training** settings, metric, stopping/selection rule and split policy.

Repeated choices based on the test use it for development; reserve new independent final data after freezing the revised procedure.

W4 Plan a new case independently

Workbook question · W4

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Teacher reasoning · W4

Teacher explanation: **Baseline prediction**s 3 and 4, errors -6 and -8, squared errors 36 and 64, MSE 50 over two rows.

Candidate is fixed at two updates with rate 0.1 on **training** only; success means candidate validation MSE < 50. No candidate results have been supplied, so success is unknown.

W5 Preserve the reserved evidence

Workbook question · W5

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

Teacher reasoning · W5

Teacher explanation: Record the inspection and changed plan. Those rows informed selection and cannot honestly serve as untouched final **evidence**.

Freeze the revised procedure and use newly reserved independent cases; retain the earlier result as development **evidence**.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires **baseline** MSE 50, all fixed settings, a strict < 50 criterion and no premature success claim. Fresh retry: use validation (3,12) and (4,16), same **baseline** and candidate settings.

Baseline errors -9 and -12 give MSE (81+144)/2 = 112.5; require a < 112.5 criterion and an untested conclusion.

Assess independent transfer

Independent question · withhold the key

Selected case: A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: **Baseline prediction**s 3 and 4, errors -6 and -8, squared errors 36 and 64, MSE 50 over two rows.

Candidate is fixed at two updates with rate 0.1 on **training** only; success means candidate validation MSE < 50. No candidate results have been supplied, so success is unknown.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect the project, support and next lesson

> Connect Capstone hypothesis and baseline to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Week 34 runs controlled learning-rate comparisons while preserving the recorded data roles.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/34, ai-5/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Capstone hypothesis and baseline lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Write a testable capstone **hypothesis** and a credible **baseline**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Supported practice answer

Reasoned practice answer: Teacher explanation: **Prediction**s 3 and 4; errors -3 and -4; squared errors 9 and 16; MSE 12.5 over two rows.

Check every learner before discussion: Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A **hypothesis** states an expectation before the comparison. Results may support or contradict it.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Give only B the eight evaluation labels during **training**. Require a **prediction** and a reason before comparing. Expected result: The proposed comparison becomes contaminated rather than a valid held-out test.

Reason: One version has gained privileged information from the reserved evaluation set.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

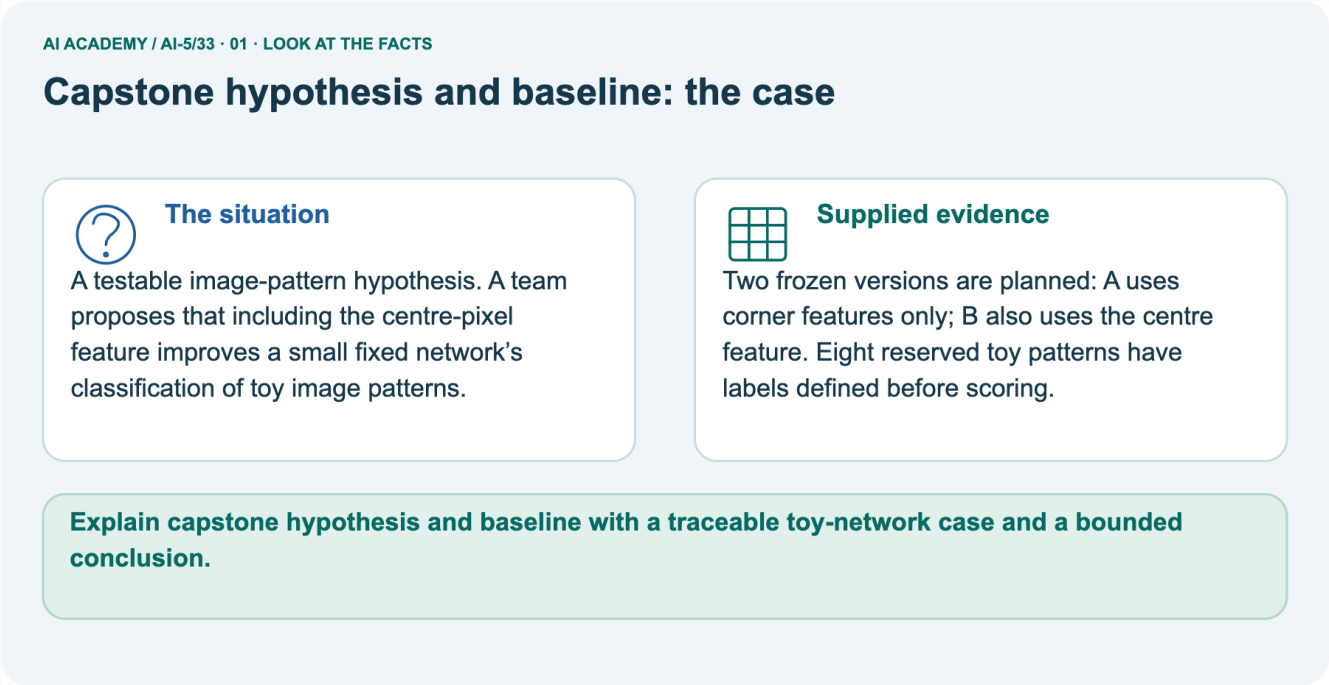
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Capstone hypothesis and baseline graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone hypothesis and baseline: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

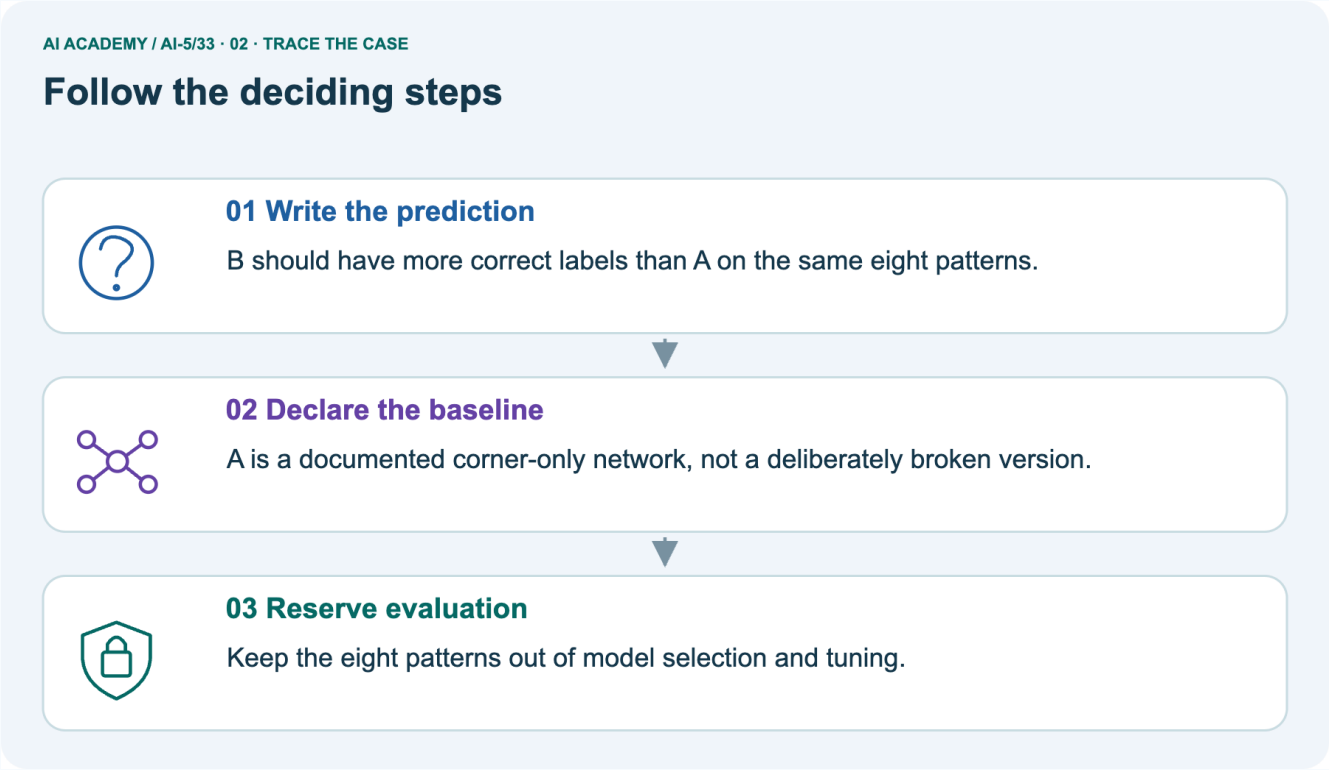
Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

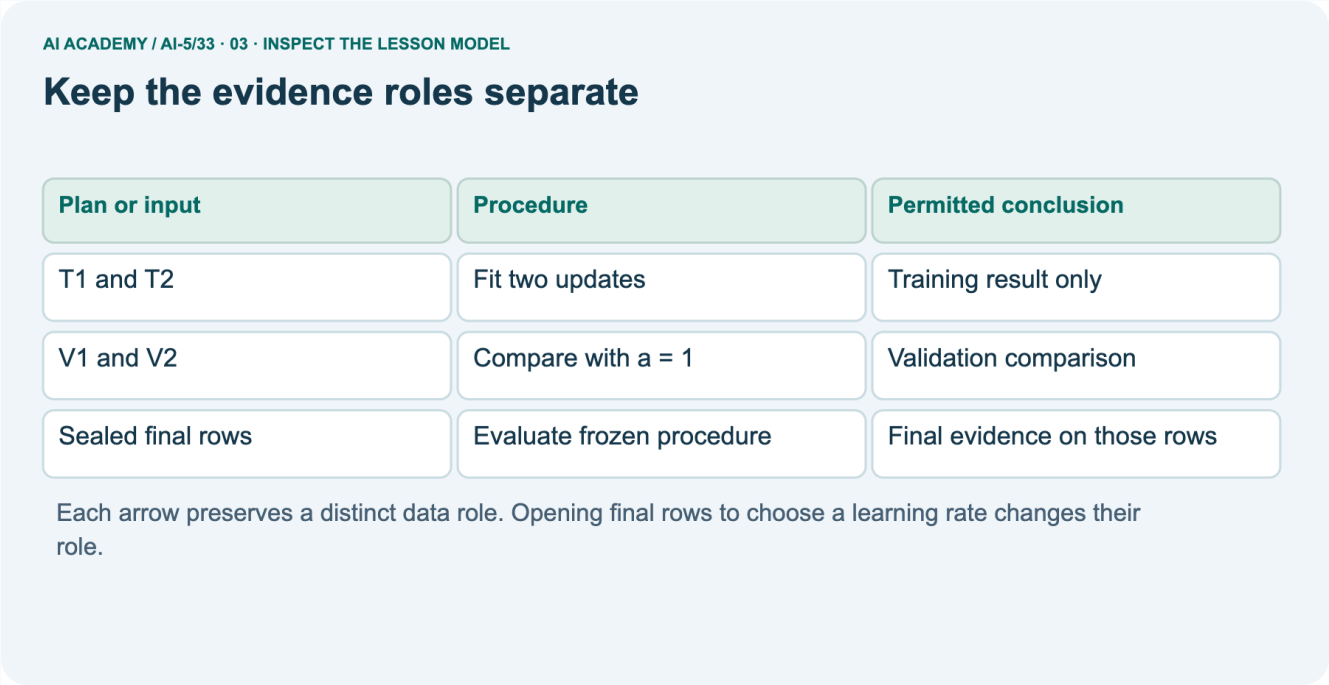
Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep the evidence roles separate



Follow the labelled picture with the matching explanation. Open the vector diagram

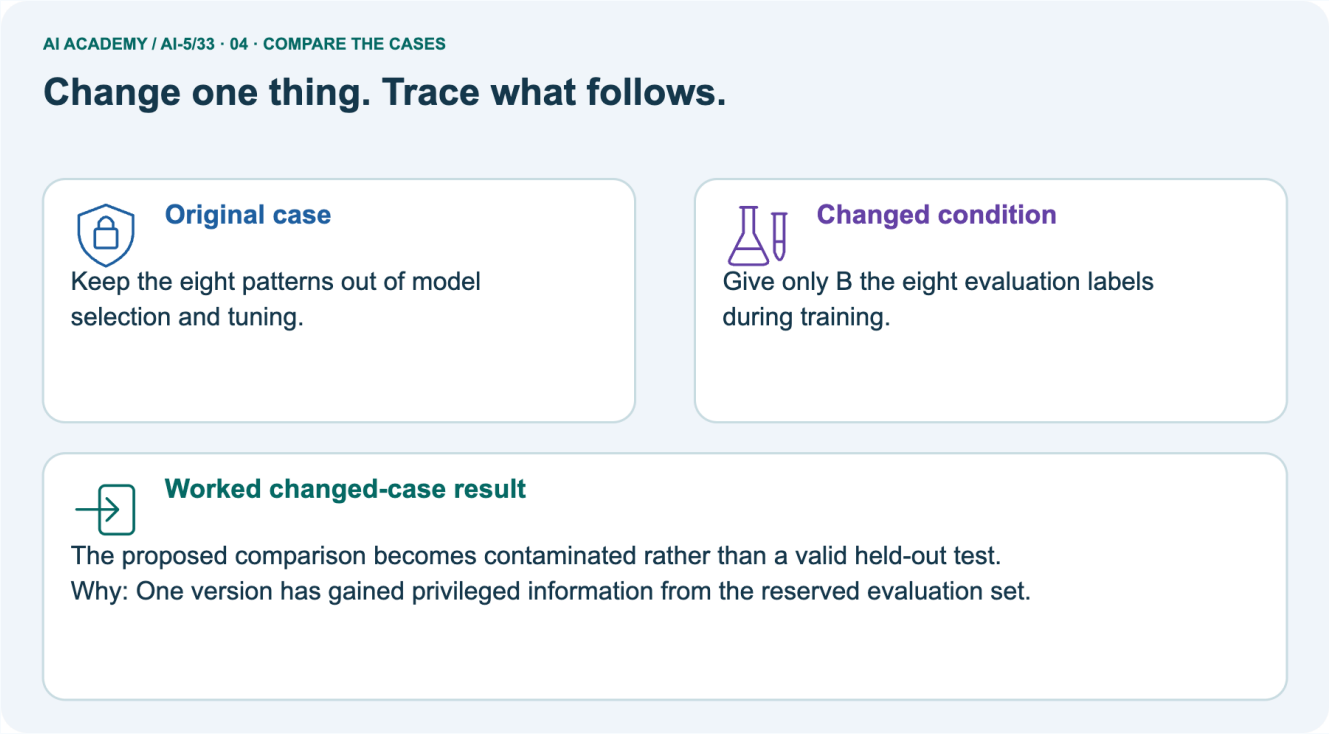
Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

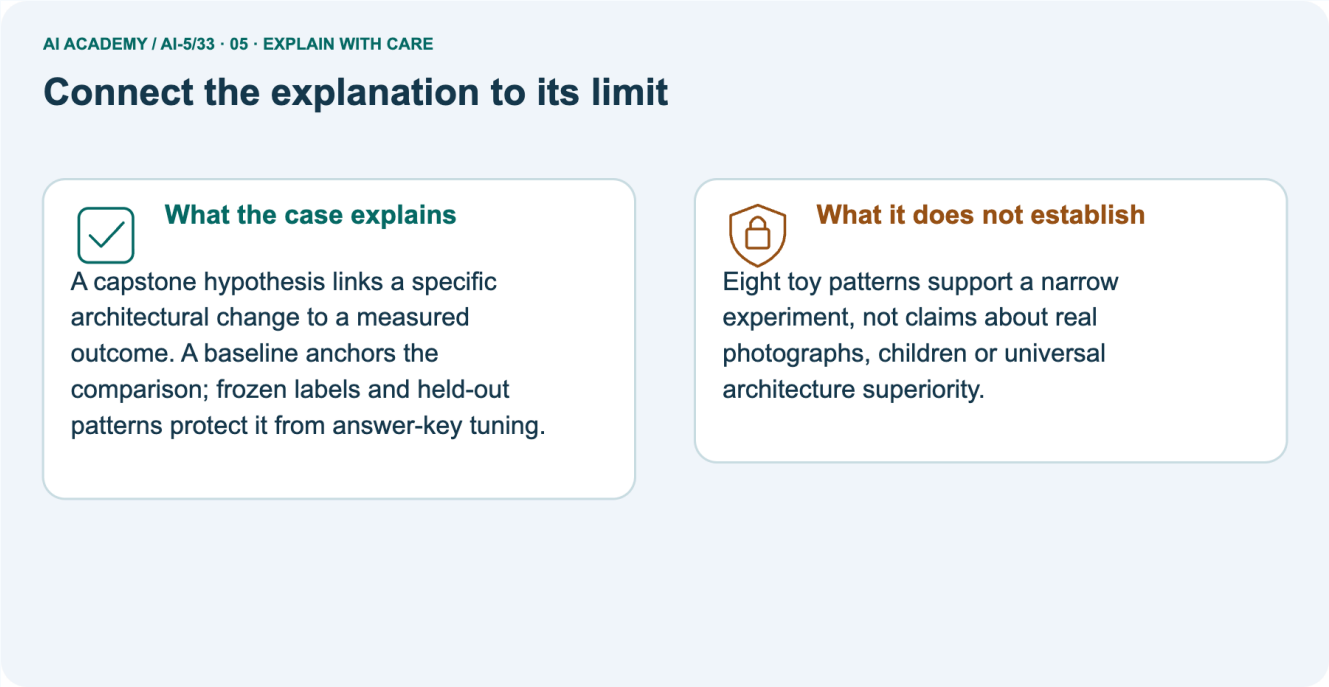
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One version has gained privileged information from the reserved evaluation set.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

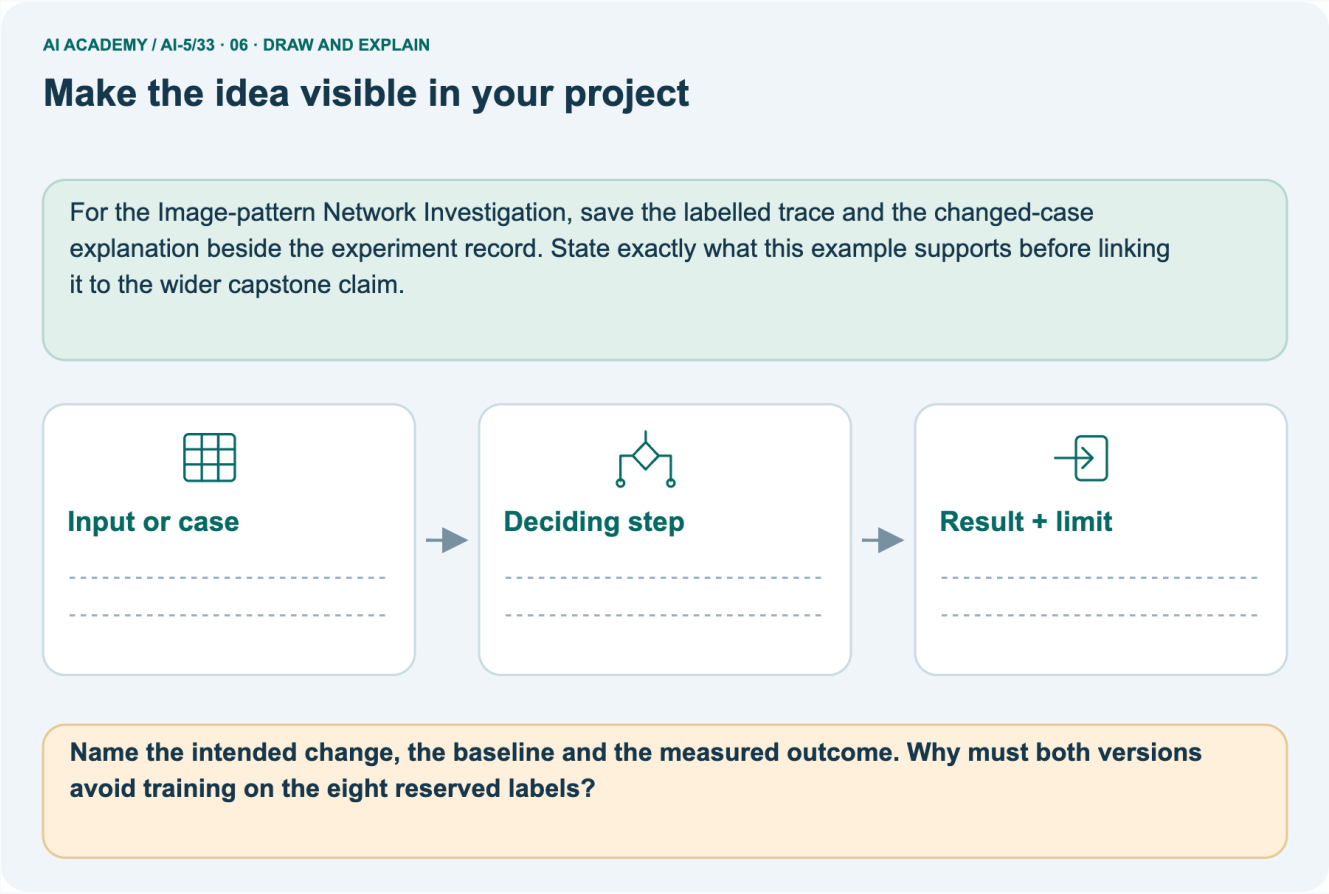
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The change is adding the centre feature; A is corner-only. The outcome is correct labels out of the same eight patterns.

Using reserved labels to train B leaks evaluation information and invalidates the unseen comparison.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.